PRIMATES, 18(1): 205–214, January 1977

Effects of Infant-infant Separation of Young Monkeys in a Free-Ranging Natural Environment*

SHEO D. SINGH
Meerut Universirty

ABSTRACT. Five infant rhesus monkeys reared togather in free-ranging natural environment for about eight months were subjected to peer separation. Separation of an infant was inflicted by way of removing all the other infants from their home range. The infants so separated showed exceedingly high levels of locomotion and vocalizations in the beginning of separation phase, but their such excited behaviour did not last for more than a day, and during the remaining separation period they appeared to be quite depressed as mainly indicated by their reduced range of locomotion, reduced motivation for food and water, and reduced level of vocalization and environmental exploration. Thus, the animals showed a biphasic response to separation, which was characterized by an initial phase of 'protest' followed by a 'despair' stage, basically similar to what has been reported in infant rhesus monkeys subjected to peer or mother separation under laboratory conditions.

INTRODUCTION

Of the several factors, experience of separation from an object of emotional attachment has been considered to be of critical importance in the aetiology of depression or psychopathological states in human beings. Initial support for such a notion has come from clinical observations of young children by SPITZ (1945, 1946). SPITZ observed that children upon separation from their mothers via hospitalization displayed a behavioural syndrome characterized by an initial phase of agitated distress ("protest") followed by extended withdrawal and passivity ("despair"), a reaction termed as "anaclitic depression". Later, similar signs of distress and depression were also observed by BOWLBY (1960, 1969) in children when separated from their mothers. BOWLBY also reported that some of the separated children showed indifference and/or hostility ("detachment") following maternal reunion.

In recent years, because of some unsurmountable difficulties of experimental work with human subjects, a number of behavioural scientists have resorted to the use of nonhuman primates in their efforts to understand human separation reactions. In all the laboratory studies done so far infant monkeys have been found to exhibit reactions strikingly similar to those of human infants and children upon separation either from their mothers (e.g., HINDE & SPENCER-BOOTH, 1971; JENSEN & TOLEMAN, 1962; KAUFMAN & ROSENBLUM, 1967; SEAY, HENSEN, & HARLOW, 1962; SEAY & HARLOW, 1963) or from peers (e.g., SUOMI, HARLOW, & DOMEK, 1970; SUOMI, 1973). Like the human infants and children the monkey infants exhibit protest despair syndrome,

*The main project of which this research was a part is being financed by Indian Council of Social Science Research (ICSSR).

with protest characterized by exceedingly high levels of activity and vocalization, and despair characterized by inactivity and withdrawal from the surrounding environment. In view of such findings a strong belief is current that separation of monkey infants from their objects of emotional attachment either mother or peers provides an excellent model for testing some of the theories about anxiety (agitated distress) and depression states in human beings (eg., KAUFMAN, 1970; MCKINNEY & BUNNY, 1969; MCKINNEY, SUOMI, & HARLOW, 1970; BECKER, 1974).

It has been found that severity and persistence of separation reactions depend very much on the nature of pre-separation and separation environments (BRANDT, BAYSINGER, & MITCHELL, 1972; SUOMI, 1973). It is apparent that laboratory environment is far removed from that obtaining in the wild at least for monkeys. In contrast to the monotonous, movement-restraining, and predator-free laboratory conditions, the natural environment allows considerable freedom of movement, a variety of environmental exposure, and is often predator-infested. Laboratory environment may not be potent enough to effect the quality of behaviour as suggested by BERNSTEIN (1967), KUMMER and KURT (1965), HALL and MAYER (1967), and ROWELL (1967), but has been found to effect considerable quantitative differences in the feral behaviour of nonhuman primates (ALLEXANDER & BOWERS, 1968; KOLLAR, EDGERTON, & BECKWITH, 1968; ROWELL, 1967). These considerations emphasize the need to study the separation effects in monkeys living in free-ranging natural environments. In addition to testing the generality of laboratory-based information such an effort would provide more relevant data for a more comprehensive understanding of depressivelike states in humans.

We, therefore, have launched a major research programme, mainly concerned with the evaluation of the effects of various types of separation in monkeys living in their natural habitats. The present study is a part of this programme, and is specifically concerned with the assessment of infant-infant separation effects. In brief, infant monkeys were reared together, without mothers, in a forest environment until the age of about eight months, then subjected to single separation according to a well-defined procedure and schedule, with behaviours being observed before, during, and after separation.

METHOD

SUBJECTS

The Ss were five rhesus monkey infants, two males and three females. They were born between April 16 and May 8 in 1973 in one of our provisionized groups of monkeys living in Asarori range of Dehradun forests. A detailed description of their habitat is given by LINDBURG (1971). In brief, the predominant vegitation in this region of the forest consists of alluvial *sal, Shorea robusta.* There are also several varieties of wild fruits such as figs, berries, and *jamum* (*Syzyquim cumini*). In the region, there are several water holes and two springs. These infants were caught along with their mothers, at which time they ranged in age from 15 to 34 days. Immediately after their capture they were separated from their mothers and housed in groups of two or three in small cages measuring $1\text{-}1/2 \times 1 \times 1$ ft and fitted with feeder holders. These cages

were perched side by side on a tree in the forest at a locale having no forest monkey groups around. After two days the cages were left open, which always remained so thereafter, thus allowing full freedom to the infants to move around and explore the surrounding environment.

After their capture they were fed *ad lib* Amulspray (Powder milk: Kaira District Co-operative Milk Producer's Union Ltd., India) at the rate of four times a day for the first two months, then three times a day for the next two months, and thereafter twice a day until the completion of this study. In addition, they were daily given an adequate amount of monkey chow (Hindustan Levers Ltd., India) and some apple or orange. Water was always available. The infants were between 222 and 242 days of age when the observations for the present study began.

Procedure

The experiment consisted of three different phases. (1) Pre-separation phase: The infants continued to live with their age mates in the group, and baseline social and individual data were gathered by observing each infant for 10 min period per day for six days from 15 to 20th of December, 1973. (2) Separation Phase: Separations began on December 21. Each of the infants was subjected to separation according to a randomly predetermined order. Separation of an infant was accomplished as follows: Frist, all the infants were captured with the help of a drope door-trapping cage, then except the one to be separated all others were housed in a standard-size laboratory cage, and removed to distance of about 1 km from the core area, and thereafter the infant held was released free to live there all alone for a period of four days without having any opportunity of any kind of interaction with its other associates. Immediately after its release the infant was observed for one full hour, and the same day-time observation schedule was followed on the remaining three days of separation period. (3) Reunion Phase: In the midforenoon of the fifth day, all the other infants were brought back, and released free in their core area, and allowed to stay in there undisturbed for two days. Immediately after the release of other infants, the infant who had already undergone separation was observed for one hour, and also for one hour on the following day at the same time of the day. Thus, one infant was separated on December 21, and reunited on December 25; one infant was separated on December 27 and reunited on December 31; one infant was separated on January 2, and reunited on January 6; one infant was separated on January 8, and reunited on January 12; and one infant was separated on January 14, and reunited on January 18. The procedure of separation and reunion remained the same for all the animals.

During each of these phases, observations were conducted around midforenoon, and the observation procedure remained as follows: In an observation session, the observer maintained continuous visual contact with the animal under observation, and recorded vocalizations in terms of frequency, and other behaviours in terms of their occurrence during 20-second intervals. The beginning of a 20-second interval was signalled by a battery-operated buzzer. The definitions of behaviour categories used in the study are given in Table 1.

To summarize, the Ss were observed for six 10-min sessions, one session per day, during the pre-separation phase; for four 60-min sessions, one session per day during

Table 1. Behavioural categories with definitions.

Locomotion:	Any movement of S inducing change in its physical location.
Passivity:	Indulging in no activity.
Environmental Explore:	Any type of manual and/or oral manipulation of inanimate objects such as tree branches, leaves, picking-up small pieces of wood.
Social Exploration:	Oral and/or manual examination and manipulation of other S, such as licking and grooming.
Social Contact:	Any type of body contact with other S other than ventral clinging.
Social Play:	Any kind of play activity directed toward other S, such as rough-and-tumble activity, mutual and nonmutual chase.
Social Sex:	All sexual behaviours-mounting, presentation and thursting whether appropriate or inappropriate.
Self-manipulation:	Self-licking, self-scratching, and self-grooming.
Self-play:	Highly active running and/or jumping without orientation, and rolling over on the ground and sliding on tree trunks.
Non-social Sex:	Oral or manual self-manipulation of genital organs.
Self-clasp:	Clasping any part of own body with hand(s) or foot (feet).
Self-bite:	Biting own body parts vigorously.
Huddle:	Self-enclosed, accompanied with lowered head.
Rock:	Repetitive nonlocomotive forward and backward movement.
Stereotyped movement:	Identical movements executed in a repetitive fashion for more than twice.
Vocalization:	Coo, screech, and any other sound emmitted.
Threat:	Facial expressions consisting of ears pulled back with lower jaw pulled down and out.
Aggression:	Any act causing physical insult to another S, such as biting, grabbing and pulling the fur, clasping, stricking, and pushing, accompanied by one or more of threat, barking, piloerection, or strutting.
Ventral-cling:	Contact of own ventral body surface with that of another individual.

the separation phase; and for two 60-min sessions, one session per day, during the reunion phase.

For each S individual behaviour category scores were summed up for the pre-separation phase, and also for the reunion phase, leaving the scores of individual observational sessions during the separation phase as such. These scores then were made comparable by converting the interval scores into percent scores (percentages of 20-second intervals in which a behaviour occurred), and by calculating the mean of the frequency scores summed over two sessions during the reunion phase. Thereafter, the obtained scores of those behaviours which occurred frequently were subjected to a one-way repeated measures analysis of variance, with time block (three experimental conditions in a successive order) as the repeated measure. The means within those categories where F ratios were significant, were compared by Duncan Range Test.

RESULTS

From among the several behaviours listed in Table 1, only four behavioural categories–locomotion, vocalization, environmental exploration, and passivity–were found to be the significant components of the separation syndrome in the present study.

In the initial stage of separation period, there occurred a significant increase ($p < .01$) in the levels of locomotion (Fig. 1), and also of vocalization (Fig. 2); in contrast,

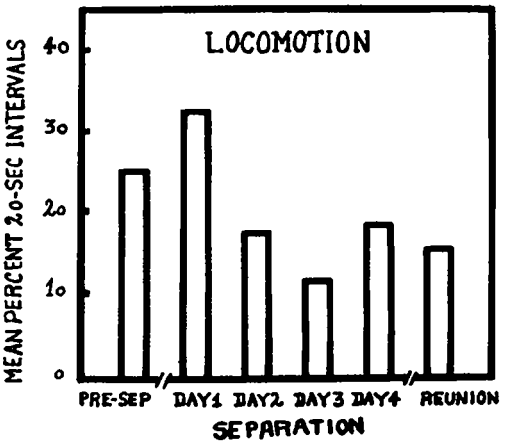


Fig. 1. Mean levels of locomotion.

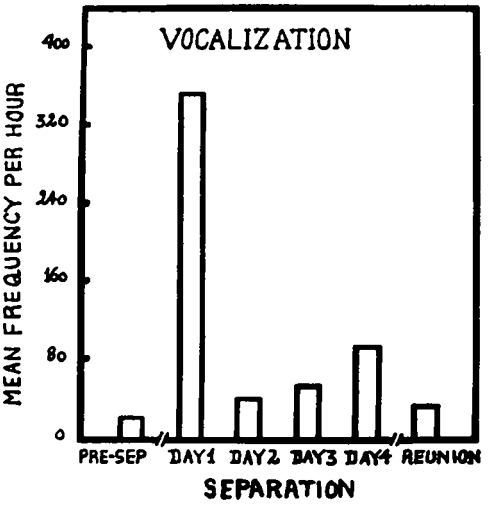


Fig. 2. Mean levels of vocalization.

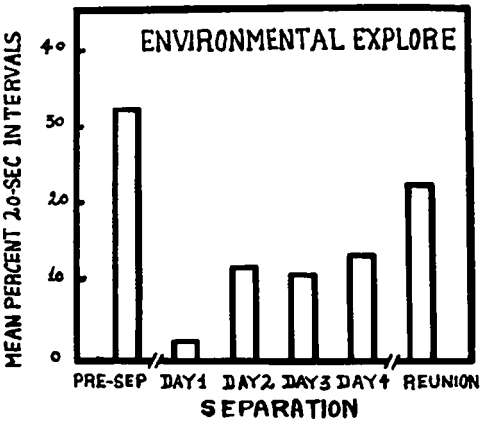


Fig. 3. Mean levels of environmental explore.

the incidence of environmental exploration (Fig. 3), defined as any tactual or oral exploration of the physical surroundings, became significantly lower than that observed during the preseparation phase or reunion phase ($p<.01$). Whatever little interaction the infants had with the physical environment during this stage of separation appeared to be rather of an aggressive nature since they were seen biting the barks of tree branches, and chewing even the nearly dry leaves rather frantically. In fact, in the initial stage, that is on the first day of separation, the behaviour of the infants was found to be predominantly characterized by hyperactivity and agitated distress; they frantically ran about, often left the core area, hurriedly climbed trees, jumped from one tree to another, and emitted high-pitched and long-drawn cries and visually scanned around while climbing on the high treetops. By and large, their behaviour gave the impression of their being in desperate search for their partners removed from the area.

Such a stage of hyperactivity and agitation did not persist long. Right on the next day of separation, except for occasional outbursts of hyperactivity and high-pitched and long-drawn cries, their locomotion and vocalization levels almost dropped to their respective baseline or reunion values; largely locomotion became slow and its range became confined to a very small portion of the core area; and they became highly inactive (Fig. 4), with a significant increase in their passivity scores over the baseline or reunion values ($p<.01$). In fact, they were observed spending a large proportion of their time sitting rather quietly underneath one or the other bush on an elevated place in the core area. They showed some increased level of envirnonmental exploration, but the increase was not of a significant order over that observed on the first day of separation. This time, however, the animals appeared to be rather gentle and cautious in exploring the physical surroundings. This state continued across the remaining days of separation.

At no stage during the separation period the infants engaged in behaviours such as self-clasp, self-mouth, rock, and huddle which in addition to passivity have been observed to be associated with the state of despair in several laboratory studies (e.g., SUOMI, HARLOW, & DOMEK, 1970; SUOMI, 1973). In the present study, the infants even

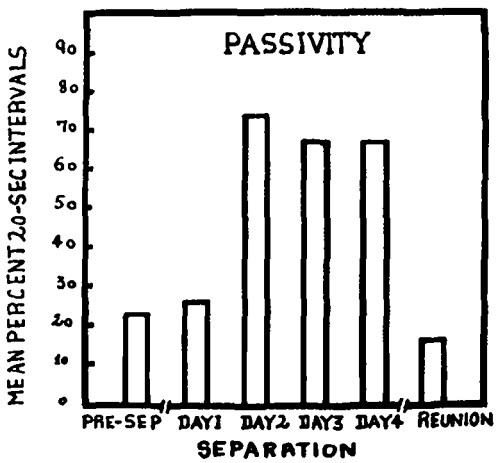


Fig. 4. Mean levels of passivity.

while sitting quiet appeared to be very vigilant and alert to any event in the surrounding environment.

It must be pointed out that during the separation phase the infants seemed to have very low motivation for food or water; they were never observed foraging for food or water, and also responded very poorly whenever called for food by the care-taking personnel.

Upon reunion, they became as active as they were during the preseparation phase, and their interactions with others were also found to be normal when compared with the baseline values. During the entire period of reunion (two days) there did not occur even a single incidence of ventral clinging, a behaviour often observed in laboratory studies (e.g., SUOMI, HARLOW, & DOMEK, 1970; SUOMI, 1973).

DISCUSSION

By and large, the findings of this study provide strong support to those conducted in laboratory settings (e.g., SUOMI, HARLOW, & DOMEK, 1970; SUOMI, 1973). In essence, the reactions of the monkey infants following separation in the free-ranging natural environment from their peer group were found to be similar to those of their counterparts separated under laboratory conditions; they exhibited a biphasic response characterized by an initial "protest" stage and a subsequent "despair" stage. The "protest" phase which ensued immediately after separation, was mainly characterised by exceedingly high levels of vocalizations which consisted mainly of high-pitched and long-drawn cries indicative of anxiety and agitated distress, and also by disoriented running about, and climbing trees in a very frantic manner. The "despair" stage which ensued right on the next day of separation and continued throughout the separation period, was mainly characterized by increased passivity, reduced range of locomotion, and reduced level of motivation for food. The "despair" state, however, was not judged to be as intense and incapicitating as has often been reported for infant monkeys following separation under laboratory conditions, since none of our infants showed any behavioural patterns like huddling, crouching, rocking, self-clasp, or self-mouth, which have often been found by laboratory workers to be associated with greater degree of depression.

The occurrence of despair reactions in a less intense form in this study cannot be attributed to any kind of instability in social bonds among the group members, since prior to starting separation they had lived long enough with each other, and lived in an environment in which the group's viability depended very much upon its cohesiveness for gathering food, and protection against ground and aerial predators. And such a mild form of depression cannot also be accounted for in terms of age, since in another study (under preparation) we have observed signs of depression even in adult monkeys following separation from their group members. To us, the environment in which the infants lived during separation seems to be the most potential factor. The forest environment in which these animals lived during their period of separation was not only stimulating, but was also predator-infested, and require a good deal of vigilance and alertness on the part of the infants for their survival, thus the environment may not be as conducive to the production of debilitating psychopathological

state as the monotonous, restrictive, and predator-free environment of the laboratory. The findings of a recent study by SUOMI (1973) also support such a view; SUOMI observed that infants separated in a vertical chamber, a structure which has been referred to as "depression sink" (SUOMI & HARLOW, 1969) showed higher levels of depression than their counterparts separated in cages. In another study, HARLOW and SUOMI (1971) also observed that just the confinement of socially naive rhesus monkeys of 6 to 13 months of age for a period of 30 days in the vertical chamber was enough to produce a psychopathological state which was similar to that resulting from separation either from mother or from peers.

In contrast to the laboratory findings, our infants did not show any detectable change in their behaviours during the reunion phase; there was not observed even a single incidence of ventral or any other kind of clinging behaviour as has often been seen in laboratory studies (e.g., SUOMI, HARLOW, & DOMEK, 1970; SUOMI, 1973). To highlight the possible interpretation of these discrepant findings, I would like to narrate an unfortunate accident to which our so-raised infant monkeys of another peer group once became the victims. One day, a wicked fellow trapped all the six infants of this group from their core area in the forest, and thereafter he sold all these infants to different persons in Dehradun city. After a hectic search we got hold of this man, and retrieved the infants after three days. Following their retrieval they were housed in one cage. To our great surprise we found all the infants clinging to each other; they clung to each other so strongly that it became difficult to separate them even after an hour, and we were able to lift all the six infants just by holding any one of them. They literally formed a round-shaped ball which we could roll on the floor. But their clinging behaviour disappeared immediately after their release in their core area in the forest. These observations suggest that the occurrence of clinging behaviour observed in laboratory studies may have been the result of the simultaneous separation-reunion procedure and/or restricted reunion conditions used by the laboratory workers. The basis for such an inference is that if separation at all induces in an individual the tendency to cling then there is a greater probability of the occurrence of clinging behaviour during reunion if all the members of the group have undergone separation simultaneously than when only one of them has undergone separation; and such a tendency is likely to incerease under anxiety which may be presumed to arise in a situation providing no opportunity for their escape from being caught and separated again.

Some speculations have been made in regard to the adaptive functions of the distress reactions to social separation (ENGEL, 1962; KAUFMAN & ROSENBLUM, 1967). According to KAUFMAN and ROSENBLUM, the initial protest state is designed to seek and summon assistance; prolongation of such a state, however, would exhaust the infant and would also expose it to predators. The despair state which is characterized by passivity and withdrawal is designed to reduce physical exhaustion and the infant's chances of being exposed to its natural enemies. Our observations of the infants during separation phase provide some empirical support to KAUFMAN and ROSENBLUM's notion. For example, within a few hours of separation, that is during the protest phase we observed several hawks hovering around the core area, and some of the infants were also attacked by these hawks, but these predators were not seen when the infants

went into the despair stage in which they spent most of their time sitting underneath one or the other bush in the core area.

In general, the results of this study extend the generality of laboratory-discovered findings in regard to the infant-infant separation effects, and in addition emphasize the role of separation-reunion procedures and mainly of the environment in which infants live during separation and reunion phases in the production of depression-like states in young rhesus monkeys.

REFERENCES

ALLEXANDER, B. K. & J. M. BOWERS, 1968. The social structure of the oregon troop of Japanese macaques. *Primates*, 8: 333–340.

BECKER, J., 1974. *Depression: Theory and Research.* Winston & Sons, Washington.

BERNSTEIN, I. S., 1967. A field study of the pigtailed monkey. (*M. nemestrina*). *Primates*, 8: 217–228.

BOWLBY, J., 1960. Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child*, 15: 9–52.

————, 1969. *Attachment and Loss*, Vol. 1. Basic Books, New York.

BRANDT, E. M., C. BAYSINGER, & R. G. MITCHELL, 1972. Separation from rearing environment in mother-reared and isolation-reared rhesus monkeys (*Macaca mulatta*). *Int. J. Psychobiol.*, 2: 193–204.

ENGEL, G. L., 1962. *Psychological Development in Health and Disease.* W. B. Saunders, Philadelphia.

HALL, K. R. L. & B. MAYER, 1967. Social interactions in a group of captive patas monkeys (*Erythrocebus patas*). *Folia primat.*, 5: 213–236.

HARLOW, H. F. & S. J. SUOMI, 1971. Production of depressive behaviours in young monkeys. *J. Autism Childhood Schizophrenia*, 1: 246–255.

HINDE, R. A. & Y. SPENCER-BOOTH, 1971. Effects of brief separations from mother in rhesus monkeys. *Science*, 173: 111–118.

JENSEN, G. D. & C. W. TOLEMAN, 1962. Mother-infant relationship in the monkey, *Macaca nemestrina:* The effect of brief separation and mother-infant specificity. *J. Comp. Physiol.*, 55: 131–136.

KAUFMAN, I. C., 1970. Mother-infant separation in monkeys: An experimental model. *Paper presented as a part of Symposium in separation and depression: Clinical and Research Aspects, Animal Behavior Society and AAAS Meeting, Chicago, December, 1970.*

———— & L. A. ROSENBLUM, 1967. The reaction to separation in infant monkeys: Anaclitic depression and conservation withdrawal. *Psychosomatic Medicine*, 29: 648–675.

KOLLAR, E. J., R. B. EDGERTON, & W. C. BECKWITH, 1968. An evaluation of the ARL colony of chimpanzees. *Archiv. General Psychiat.*, 19: 580–595.

KUMMER, H. & F. KURT, 1965. A comparison of social behavior in captive and wild hamadryas baboons. In: *The Baboon in Medical Research*, H. VEGTBORG (ed.), University of Texas Press, pp. 65–80.

LINDBURG, D. G., 1971. The rhesus monkey in North India: An ecological and behavioral study. In: *Primate Behavior*, Vol. II, L. A. ROSENBLUM (ed.), Academic Press, New York & London, pp. 1–106.

MCKINNEY, W. T. & W. E. BUNNY, 1969. Animal model of depression. *Archiv. Gen. Psychiat.*, 21: 240–248.

————, S. J. SUOMI, & H. F. HARLOW, 1970. New models of separation and depression in rhesus monkeys. *Paper presented as a part of the symposium on separation and depression: Clinical and Research Aspects; Animal Behavior Society and AAAS Meeting, Chicago, December, 1970.*

ROWELL, T. E., 1967. A quantitative comparison of the behavior of a wild and caged baboon group. *Animal Behav.*, 15: 499–509.

SEAY, B., E. W. HANSEN, & H. F. HARLOW, 1962. Mother-infant separation in monkeys. *J. Child psychol. Psychiat.*, 3: 123–132.

——— & H. F. HARLOW, 1965. Maternal separation in the rhesus monkey. *J. Nerv. Mental Diseas.*, 140: 434–440.

SPITZ, R. H., 1945. *Hospitalism: An Inquiry into the Genesis of Psychiatric Conditions in Early Childhood.* International Universities Press, New York.

———, 1946. Anaclitic depression. *Psychoanalyt. Study Child.*, 2: 313–341.

SUOMI, S. J., 1973. Repetitive peer separation of young monkeys: Effects of vertical chamber confinement during separations. *J. Abnorm. Psychol.*, 81: 1–10.

——— & H. F. HARLOW, 1969. Apparatus conceptualization for psychopathological research in monkeys. *Behav. Res. Methods Instrument.*, 1: 247–250.

———, ———, & C. J. DOMEK, 1970. Effect of repetitive infant-infant separation of young monkeys. *J. Abnorm. psychol.*, 76: 161–172.

—Received *June 4, 1975*; Accepted *September 27, 1975*

Author's Address: SHEO D. SINGH, *Primate Research Laboratory, Meerut University, Meerut 250001, India.*